

## QUANTIZER REPORT SEMANTICS

# Three counts, three different questions

Observed labels are reachable; reachable labels are nominal. Population support is separate.

### Nominal joint labels

Requested combinatorial label space

$$N = B^D$$

### Finite-binary64 reachable labels

At least one finite represented input maps to each label

$$R = \text{product } r_d$$

### Observed joint tuples

Present in the evaluated rows

$$O$$

$$O \leq R \leq N$$

$$N - O = (N - R) + (R - O)$$

### SEPARATE OBJECT

### Population support

Positive probability under a declared population law

Not implied by N, R, or O

### REPORT DIFFERENCES

N - R structurally unreachable

R - O reachable but unobserved

N - O legacy empty-cell field

None means only  
u128 product overflow.

DO NOT COMPACT SILENTLY: REPEATED EDGES ARE PART OF THE FITTED MAP AND ITS PROVENANCE